

Project Progress

Current Status

Current Phase:

Document Extraction & Flattening

The core document ingestion pipeline has been completed.

Completed Work

Environment Setup

Completed

- Virtual environment
 - Dependency installation
 - PyMuPDF integration
 - Project structure
-

Digital PDF Extractor

Completed

Implemented

DigitalPDFExtractor

Responsibilities

- Open PDF
- Read pages
- Extract blocks
- Extract lines
- Extract spans
- Preserve formatting metadata
- Build structured document

Uses

```
page.get_text("dict")
```

instead of word-based extraction.

Data Models

Completed

Implemented

- Document
- Page
- TextBlock
- Line
- Span
- TextElement

Hierarchy

Document

↓

Page

↓

TextBlock

↓

Line

↓

Span

Flattening Module

Completed

Implemented

```
DocumentFlattener
```

Converts

```
Document
```

into

```
List[TextElement]
```

Each element contains

- unique id
- page
- block
- line
- text
- bbox

This is the primary input for future comparison modules.

Validation

Successfully tested using sample PDFs.

Current output

```
45 flattened text elements
```

Extraction order has been verified.

Bounding boxes preserved.

Page hierarchy preserved.

Architectural Decisions

Completed

- Block-based extraction
 - Line-based comparison units
 - Immutable extraction output
 - Separate flattening stage
 - Modular project structure
-

Remaining Work

Immediate Next Step

Sequence Alignment

Purpose

Correctly match corresponding lines before comparison.

This prevents a single insertion or deletion from causing cascading mismatches throughout the document.

Expected output

```
AlignedPair[]
```

After Alignment

Normalization

Responsibilities

- Whitespace normalization
- Unicode normalization
- Hidden character detection
- Case normalization

Original text will always remain preserved.

Then

Text Comparator

Will detect

- Character changes
- Word changes
- Sentence changes
- Insertions
- Deletions
- Spelling mistakes
- Hidden Unicode differences

Future Milestones

Formatting comparison

Image OCR

Scanned PDF OCR

Table comparison

Layout comparison

Visual report generation

Interactive HTML report

Current Pipeline

PDF

↓

DigitalPDFExtractor

↓

Document

↓

DocumentFlattener

↓

TextElement[]

This pipeline has been completed and verified.

Immediate Objective

Implement the Sequence Aligner.

The aligner will map corresponding TextElements between two documents before any normalization or comparison takes place.

This alignment layer will become the foundation for every future comparison module, including text, formatting, OCR, and image comparison.